

Stage 1 — real terrain (the Forge field)

Decision: the field is a **real festival venue that SEED services**, not invented terrain. SEED's portfolio includes **Lightning in a Bottle**, so the field is a real LIB site.

Chosen site: Lake San Antonio, Bradley, CA

(Monterey County) — LIB's historic home, the rugged "California Hills of Bradley." Oak-savanna hills around a reservoir → real elevation, real chokepoints. *Approx. centre:* `35.80° N, 120.88° W`. Demo bbox: a small ~2–4 km window (start small → scale later).

Swap option: Buena Vista Lake, Kern County, CA (`35.19° N, 119.32° W`) — LIB's *current* venue (2022–present). Flatter; use if "where they are now" beats "best terrain."

chitin-coast is **not** a data source here. It is referenced only as evidence of the skill ("I've built DEM/terrain pipelines with rasterio"). The data is real.

Why real data is a feature, not decoration

The **lake and creeks at the venue become the "ravine"** that bridges cross. The chokepoints aren't

hand-drawn — they fall out of the real DEM. That is the "bend the mechanic to the tool/data" principle made literal.

Pipeline (uses real tools, nothing hand-rolled)

1. **Pick bbox** around the venue (small for the demo).
2. **Fetch DEM** — open data, no auth:
 - primary: **USGS 3DEP (10 m)** via OpenTopography API
 - easy fallback: **AWS Terrain Tiles** (Terrarium PNG → decode to elevation)
1. **Read it** with `rasterio` (the chitin-coast stack).
2. **Cost surface:** slope from elevation → traversal cost; water + too-steep =

impassable → these are the natural **ravine / chokepoint** edges.

1. **Place sectors:** sample *N* control points on traversable land → Voronoi regions = sectors (use `scipy/networkx`, not custom code).

1. **Build the graph:** neighbouring sectors connected; ****edge cost = least-cost path over real terrain (true travel-time); edges crossing water/steep ground are gated**** (need a bridge).

1. **Export `field.json`** (sectors + edges + costs + gated flags) → consumed by

BOTH the Python RL env and the web demo. One field, one source of truth.

Scale story (your "start small → prove it scales")

Small bbox + ~12–30 sectors + 3v3 now → the *same* pipeline takes a bigger bbox, finer DEM, and more

sectors/agents with more compute. Nothing in the pipeline is hardcoded to the small size — that's the scalability proof for an employer.